Introduction to Quantum Computing

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1 Basics concepts

Classical computing is based on the concept of bits, which can take values of either 0 or 1. **Quantum bits**, or **qubits**, on the other hand, can exist in a superposition of 0 and 1.

A quantum state $|\psi\rangle$ is described as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The ket notation of $|\psi\rangle$ stands for

$$|\psi\rangle \in \mathbb{C}^2, \quad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \implies |\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Difference from Classical Bit Unlike a classical bit, a qubit cannot be directly observed: the amplitudes α and β cannot be measured. A standard measurement will result in either 0 or 1, with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.

Additionally, the measurement in general **changes** the qubit state ("wavefunction collapse"). For example, if we measure $|\psi\rangle$ and get 0, the state collapses to $|0\rangle$, and if we get 1, it collapses to $|1\rangle$.

In practise we can estimate the probabilities $|\alpha|^2$ and $|\beta|^2$ by repeating the same experiment many times (outcome statistics). These repetitions are called **shots** or **trials**.

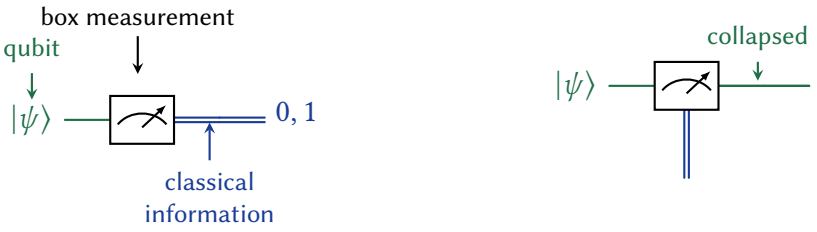


Figure 1: Circuit notation for a measurement.

What is a qubit physically? Many poossible realizations of qubits are>

- Two different polarizations of a photon (e.g.,left/right circular)
- Alignment of nuclear or electron spin ↑↓
- Ground state or excited state of an atom (electronic state)
- Clockwise or counterclockwise current in a superconducting loop with Josephson junctions (flux qubits)

Useful graphical representations of a qubit state are the **Bloch sphere**: if α and β happen to be real-valued that we can find angle $\theta \in \mathbb{R}$ such that $\alpha = \cos(\theta/2)$ and $\beta = \sin(\theta/2)$

$$|\alpha|^2 + |\beta|^2 = \cos^2(\theta/2) + \sin^2(\theta/2) = 1.$$

In general, α and β are complex numbers. Using so-called phase angles γ for α and $\gamma + \varphi$ for β, we represent

$$\alpha = e^{i\gamma}\cos(\theta/2), \qquad \beta = e^{i(\gamma+\varphi)}\sin(\theta/2), \quad \theta \in [0, \pi], \ \varphi \in [0, 2\pi).$$

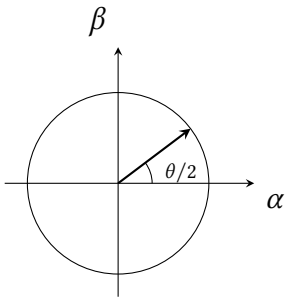


Figure 2: Real-valued α, β: the state is a point on the unit circle.

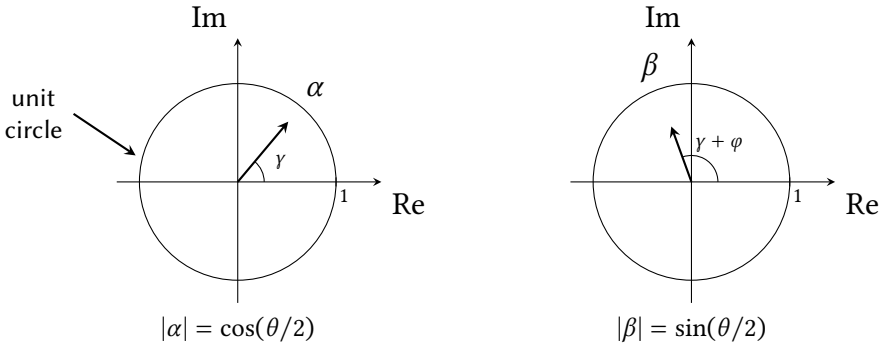


Figure 3: Complex planes of α and β.

Then

$$\begin{aligned} |\psi\rangle &= e^{i\gamma}\cos(\theta/2)|0\rangle + e^{i(\gamma+\varphi)}\sin(\theta/2)|1\rangle \\ &= e^{i\gamma}\left(\cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle\right) \end{aligned}$$

thus, up to the global phase $e^{i\gamma}$ (which is not observable), $|\psi\rangle$ is characterized by two angles θ and φ and these specify the point on the **Bloch sphere** defined as:

$$\vec{r} = \begin{pmatrix} \cos(\varphi)\sin(\theta) \\ \sin(\varphi)\sin(\theta) \\ \cos(\theta) \end{pmatrix} \in \mathbb{R}^3$$

On the surface of a sphere:

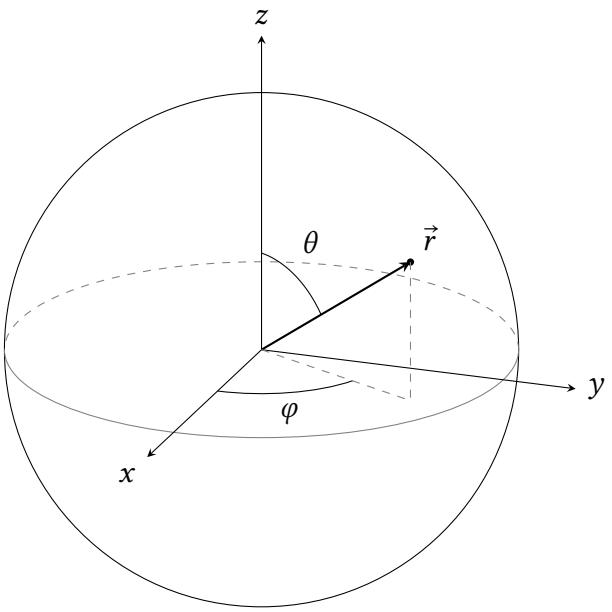


Figure 4: Bloch sphere: the point $\vec{r}$ is specified by the angles θ and φ.